Formalism implemented in `pycqed`: Fewest-switches surface hopping on non-Hermitian polaritonic surfaces built from machine-learned two-level electronic structure

PyCQED development notes (branch `refactor/ml-polariton-fssh`)

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Abstract

These notes document the equations implemented in the `pycqed` package. A molecule is described by a two-level electronic model with energies $E_g(\mathbf{R})$, $E_e(\mathbf{R})$ and dipole vectors $\boldsymbol{\mu}_{gg}(\mathbf{R})$, $\boldsymbol{\mu}_{ee}(\mathbf{R})$, $\boldsymbol{\mu}_{eg}(\mathbf{R})$, which an external (e.g. machine-learned) model supplies. The molecule couples to a single lossy cavity mode truncated at n_{f} photons. The loss enters through the complex mode frequency $\omega - i\gamma/2$, which makes the polaritonic Hamiltonian non-Hermitian but complex symmetric. We cover:

- the Jaynes–Cummings, Rabi and Pauli–Fierz Hamiltonians and the basis ordering;
- the treatment of the non-Hermitian eigenproblem;
- closed-form Jaynes–Cummings eigenpairs;
- polaritonic forces and non-adiabatic coupling vectors, including the chain-rule contribution of the intrinsic electronic coupling $\langle g|\nabla e\rangle$;
- the surface-hopping update (electronic propagation, hops, momentum rescaling, decoherence);
- photon loss treated as quantum jumps;
- Langevin nuclear dynamics;
- initial-condition sampling.

The last section lists the approximations made and the physics that is not yet included.

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1 Conventions

Atomic units are used throughout ($\hbar = m_e = e = 1$). Energies are in Hartree, lengths in bohr and times in atomic time units. The CODATA 2018 conversion $1\,E_h = 27.211386245988\,\mathrm{eV}$ is used.

- **Nuclei.** The nuclear coordinates are a flat vector $\mathbf{R} \in \mathbb{R}^{N_{\mathrm{dof}}}$, typically the $3N$ Cartesians. The momenta are $\mathbf{P}$, the masses form the diagonal matrix M, and $\mathbf{v} = \mathsf{M}^{-1}\mathbf{P}$. Products such as $\mathbf{v} \cdot \mathbf{d}$ contract over the N_{dof} nuclear components.
- **Matrices.** Matrices in the product basis are written in sans-serif (H, D, ...).
- **Polariton labels.** Polaritonic states carry capital indices J, K; the active surface is a.
- **Transpose vs. adjoint.** A superscript T is the plain transpose, without complex conjugation. A $\dagger$ is the Hermitian adjoint.

2 Electronic input

2.1 Data supplied by the electronic model

At each geometry the electronic model (`ElectronicModel.evaluate`) returns the following, in its adiabatic basis $\{|g(\mathbf{R})\rangle, |e(\mathbf{R})\rangle\}$:

$$E_a(\mathbf{R}), \quad \boldsymbol{\mu}_{ab}(\mathbf{R}) = \langle a|\hat{\boldsymbol{\mu}}|b\rangle \in \mathbb{R}^3, \quad \nabla E_a, \quad \nabla\boldsymbol{\mu}_{ab}, \quad \mathbf{d}(\mathbf{R}) \equiv \langle g|\nabla e\rangle \in \mathbb{R}^{N_{\mathrm{dof}}}, \qquad a, b \in \{g, e\}. \tag{1}$$

The dipole matrix is symmetric, $\boldsymbol{\mu}_{ge} = \boldsymbol{\mu}_{eg}$. The intrinsic non-adiabatic coupling $\mathbf{d}$ defaults to zero when the model does not provide it. The electronic derivative-coupling matrix is

$$D_{ab} = \langle a|\nabla b\rangle, \qquad D = \begin{pmatrix} 0 & \mathbf{d} \\ -\mathbf{d} & 0 \end{pmatrix}. \tag{2}$$

Missing gradients are obtained by central finite differences with step h (default 10^{-4} bohr):

$$\partial_k X \approx \frac{X(\mathbf{R} + h\hat{\mathbf{e}}_k) - X(\mathbf{R} - h\hat{\mathbf{e}}_k)}{2h}, \tag{3}$$

which costs $2N_{\mathrm{dof}}$ extra model calls. Before differencing, the sign of $\boldsymbol{\mu}_{eg}(\mathbf{R} \pm h\hat{\mathbf{e}}_k)$ is aligned with $\boldsymbol{\mu}_{eg}(\mathbf{R})$, so that a sign flip inside the model does not produce a spurious derivative.

An exact second moment $\langle a|\hat{\mu}_i\hat{\mu}_j|b\rangle$ and its gradient may also be supplied. It is used for the dipole self-energy (Sec. 4).

2.2 Projection onto the cavity polarization

Every dipole is scalarized along the unit polarization vector $\hat{\boldsymbol{\epsilon}}$ before it enters any Hamiltonian:

$$M_{ab}(\mathbf{R}) = \hat{\boldsymbol{\epsilon}} \cdot \boldsymbol{\mu}_{ab}(\mathbf{R}), \qquad \nabla M_{ab} = \hat{\boldsymbol{\epsilon}} \cdot \nabla\boldsymbol{\mu}_{ab}, \qquad Q_{ab}^{\mathrm{exact}} = \sum_{ij} \epsilon_i \epsilon_j \langle a|\hat{\mu}_i\hat{\mu}_j|b\rangle. \tag{4}$$

2.3 Phase (gauge) continuity of the excited state

The adiabatic state $|e\rangle$ is defined only up to its sign. Under $|e\rangle \to -|e\rangle$:

$$M_{ge} \to -M_{ge}, \qquad \nabla M_{ge} \to -\nabla M_{ge}, \qquad \mathbf{d} \to -\mathbf{d}, \qquad Q_{ge} \to -Q_{ge}, \tag{5}$$

while E_a, M_{gg} and M_{ee} are unchanged. Two physically identical points can therefore look discontinuous if the model returns uncorrelated signs. Along a trajectory, the data at $\mathbf{R}' = \mathbf{R} + \Delta\mathbf{R}$ are compared with the reference data at $\mathbf{R}$ through the overlap score

$$S = \boldsymbol{\mu}_{eg}(\mathbf{R}') \cdot \left[\boldsymbol{\mu}_{eg}(\mathbf{R}) + \nabla\boldsymbol{\mu}_{eg}(\mathbf{R})\,\Delta\mathbf{R}\right] + \nabla\boldsymbol{\mu}_{eg}(\mathbf{R}') : \nabla\boldsymbol{\mu}_{eg}(\mathbf{R}) + \mathbf{d}(\mathbf{R}') \cdot \mathbf{d}(\mathbf{R}), \tag{6}$$

and the phase of $|e(\mathbf{R}')\rangle$ is flipped if $S < 0$. Two features make the test robust:

- The first term uses a first-order extrapolation, so a transition dipole that genuinely passes through zero is not flagged as a flip.
- The gradient and NAC terms break the tie when $\boldsymbol{\mu}_{eg} \approx 0$.

This alignment is applied before the projection, at every step and in every surface scan (`align_phase`).

3 Cavity mode and product basis

One cavity mode has:

- real frequency ω;
- loss rate γ (the decay rate of the photon number), entering through the complex frequency

$$\tilde{\omega} = \omega - \tfrac{i}{2}\gamma; \tag{7}$$

- light–matter coupling λ and polarization $\hat{\boldsymbol{\epsilon}}$, i.e. $\boldsymbol{\lambda} = \lambda\hat{\boldsymbol{\epsilon}}$;
- Fock cutoff $n_{\mathrm{f}} \geq 1$.

The bilinear coupling prefactor is

$$\kappa = \sqrt{\frac{\omega}{2}}\,\lambda, \tag{8}$$

which deliberately uses the *real* frequency. The legacy parameterization in terms of a coupling energy g maps to $\lambda = g/(\sqrt{\omega/2}\,\mu_{\mathrm{ref}})$ (`CavityMode.from_coupling_strength`).

Basis and ordering. The working basis is the product of the moving electronic adiabats and the Fock states,

$$|a, n\rangle \equiv |a(\mathbf{R})\rangle \otimes |n\rangle, \qquad a \in \{g{=}0,\ e{=}1\}, \quad n = 0, \ldots, n_{\mathrm{f}}, \tag{9}$$

of dimension $N_s = 2(n_{\mathrm{f}} + 1)$, ordered by the index

$$k(a, n) = a\,(n_{\mathrm{f}} + 1) + n. \tag{10}$$

The first $n_{\mathrm{f}} + 1$ entries are therefore $|g, 0\rangle, \ldots, |g, n_{\mathrm{f}}\rangle$ and the next $n_{\mathrm{f}} + 1$ are $|e, 0\rangle, \ldots, |e, n_{\mathrm{f}}\rangle$.

Operators. In the truncated Fock space, $\hat{b}|n\rangle = \sqrt{n}|n-1\rangle$. On the matter side, $\hat{\sigma}_+ = |e\rangle\langle g|$, $\hat{\sigma}_- = |g\rangle\langle e|$, and we write $\hat{P}_{ab} = |a\rangle\langle b| \otimes \hat{1}$. All matrices are built with Kronecker products, matter $\otimes$ photon.

4 Hamiltonians

All three models share the form

$$\hat{H}_X(\mathbf{R}) = \sum_a E_a(\mathbf{R})\,\hat{P}_{aa} + \tilde{\omega}\;\hat{1} \otimes \hat{b}^\dagger\hat{b} + \hat{V}_X(\mathbf{R}), \tag{11}$$

where the photon zero-point energy $\omega/2$ is omitted as a constant shift. The coupling terms are

$$\text{JC:} \quad \hat{V}_{\mathrm{JC}} = \kappa\, M_{eg}(\mathbf{R})\left(\hat{\sigma}_+ \otimes \hat{b} + \hat{\sigma}_- \otimes \hat{b}^\dagger\right), \tag{12}$$

$$\text{Rabi:} \quad \hat{V}_{\mathrm{Rabi}} = \kappa\, M_{eg}(\mathbf{R})\,(\hat{\sigma}_+ + \hat{\sigma}_-) \otimes \left(\hat{b} + \hat{b}^\dagger\right), \tag{13}$$

$$\text{PF:} \quad \hat{V}_{\mathrm{PF}} = \kappa \sum_{ab} M_{ab}(\mathbf{R})\,|a\rangle\langle b| \otimes \left(\hat{b} + \hat{b}^\dagger\right) + \frac{\lambda^2}{2}\sum_{ab} Q_{ab}(\mathbf{R})\,\hat{P}_{ab}. \tag{14}$$

The Pauli–Fierz model is the length-gauge, dipole-approximation Hamiltonian $\hat{H}_{\rm el}+\omega\hat{b}^\dagger\hat{b}+\sqrt{\omega/2}\,(\boldsymbol{\lambda}\cdot\hat{\boldsymbol{\mu}})(\hat{b}+\hat{b}^\dagger)+\frac{1}{2}(\boldsymbol{\lambda}\cdot\hat{\boldsymbol{\mu}})^2$, projected onto $\{|g\rangle,|e\rangle\}$. It keeps the permanent dipoles and the dipole self-energy (DSE). The DSE matrix is either the square of the projected dipole matrix, or the exact projected second moment when the model provides one:

$$Q_{ab}=(M^2)_{ab}=\sum_{c\in\{g,e\}}M_{ac}M_{cb}\qquad\text{or}\qquad Q_{ab}=Q_{ab}^{\rm exact}. \tag{15}$$

The DSE can be switched off for testing (`dse=False`).

Writing the coupling terms with basis tensors $\hat{T}_{ab}$, $\hat{V}_X=\kappa\sum_{ab}M_{ab}\hat{T}_{ab}$ $[+\frac{\lambda^2}{2}\sum_{ab}Q_{ab}\hat{P}_{ab}]$, the three models differ only in which tensors are non-zero:

	$\hat{T}_{gg},\ \hat{T}_{ee}$	$\hat{T}_{eg}$	$\hat{T}_{ge}$
JC	0	$\hat{\sigma}_+\otimes\hat{b}$	$\hat{\sigma}_-\otimes\hat{b}^\dagger$
Rabi	0	$\hat{\sigma}_+\otimes(\hat{b}+\hat{b}^\dagger)$	$\hat{\sigma}_-\otimes(\hat{b}+\hat{b}^\dagger)$
PF	$\hat{P}_{aa}(\hat{1}\otimes(\hat{b}+\hat{b}^\dagger))$	$\hat{\sigma}_+\otimes(\hat{b}+\hat{b}^\dagger)$	$\hat{\sigma}_-\otimes(\hat{b}+\hat{b}^\dagger)$

Properties.

- Every coupling is real, so the matrix H is *complex symmetric*: $\mathsf{H}^T=\mathsf{H}$, but $\mathsf{H}^\dagger\neq\mathsf{H}$ when $\gamma>0$.
- The anti-Hermitian part is $-\frac{i}{2}\gamma\,\hat{1}\otimes\hat{b}^\dagger\hat{b}$.
- $\hat{H}_{\rm JC}$ conserves the excitation number $\hat{N}_{\rm exc}=\hat{P}_{ee}+\hat{1}\otimes\hat{b}^\dagger\hat{b}$; the Rabi and PF Hamiltonians do not.
- The legacy code corresponds to the Rabi model with $n_{\rm f}=1$ and constant $\kappa M_{eg}=g$.

4.1 Gradient of the matrix elements

Every **R**-dependent term is linear in the matter matrix elements, so

$$\nabla\mathsf{H}=\sum_a\nabla E_a\,\hat{P}_{aa}+\kappa\sum_{ab}\nabla M_{ab}\,\hat{T}_{ab}+\frac{\lambda^2}{2}\sum_{ab}\nabla Q_{ab}\,\hat{P}_{ab},\qquad\nabla(M^2)=(\nabla M)M+M(\nabla M). \tag{16}$$

Here $\nabla\mathsf{H}$ is the derivative of the matrix *elements* in the moving basis. It is not the matrix of the operator $\nabla\hat{H}$. The two are related through the basis-coupling matrix

$$\mathsf{D}=\sum_{ab}D_{ab}\hat{P}_{ab}=\begin{pmatrix}0&\mathbf{d}\\-\mathbf{d}&0\end{pmatrix}\otimes\hat{1},\qquad\langle a,n|\nabla\hat{H}|b,m\rangle=\big(\nabla\mathsf{H}-[\mathsf{H},\mathsf{D}]\big)_{an,bm}. \tag{17}$$

5 Non-Hermitian eigenproblem

5.1 c-normalized eigenvectors

Because $\mathsf{H}^T=\mathsf{H}$, the left eigenvector belonging to E_J is the transpose of the right one. The right eigenvectors $\mathsf{H}u_J=E_Ju_J$ are normalized with the bilinear ("c") product:

$$u_J^Tu_K=\delta_{JK},\qquad U=(u_1,\ldots,u_{N_s}),\qquad U_{\rm L}=U^T,\qquad U_{\rm L}\mathsf{H}U=\mathrm{diag}(E_J). \tag{18}$$

The energies E_J are complex. Their real parts define the potential-energy surfaces, and $-2\,\mathrm{Im}\,E_J$ is the photon-loss rate of state J. If the computed U^TU deviates from the identity by more than 10^{-8} (e.g. near a degeneracy), the code falls back to $U_\mathrm{L} = U^{-1}$. For $\gamma = 0$, U is real orthogonal and everything reduces to the Hermitian theory.

Exceptional points. The c-normalization fails when $u^Tu \to 0$, i.e. at an exceptional point (EP), where two eigenpairs coalesce. Before normalization the code evaluates

$$r_J = \frac{|u_J^T u_J|}{u_J^\dagger u_J} \in (0, 1] \tag{19}$$

and issues an `ExceptionalPointWarning` if any $r_J < 10^{-3}$. Note that $1/r_J^2$ is the Petermann factor.

5.2 State tracking and sign gauge

At $t = 0$ the states are sorted by $\mathrm{Re}\,E_J$. At later steps the columns are permuted with the assignment π that maximizes $\sum_J |(U_\mathrm{L}^\mathrm{prev} U)_{J\pi(J)}|$ (Hungarian algorithm). Labels therefore follow the character of each state through crossings of uncoupled states rather than the energy ordering.

The c-normalization fixes each eigenvector up to a sign only. The sign is chosen so that

$$\mathrm{Re}\,(U_\mathrm{L}^\mathrm{prev} U)_{JJ} > 0 \tag{20}$$

which is a discrete parallel transport. At $t = 0$ the largest-magnitude component of each column is given a positive real part instead.

5.3 Polaritonic amplitudes and populations

For a wavefunction ψ in the product basis, the polaritonic amplitudes and the reported populations are

$$c = U_\mathrm{L}\psi, \qquad \rho_J = \frac{|c_J|^2}{\sum_K |c_K|^2}. \tag{21}$$

For $\gamma > 0$ the right eigenvectors are not orthogonal in the ordinary sense, so $\sum_K |c_K|^2 \neq \|\psi\|^2$ in general. The populations are therefore normalized explicitly.

6 Forces and non-adiabatic couplings

6.1 Chain-rule expressions

The polaritonic states are

$$|\Psi_J(\mathbf{R})\rangle = \sum_{a,n} U_{(a,n),J}(\mathbf{R})\,|a(\mathbf{R})\rangle \otimes |n\rangle. \tag{22}$$

Their $\mathbf{R}$-dependence enters both through the expansion coefficients U and through the electronic adiabats. Differentiating $\mathsf{H}U = U\,\mathrm{diag}(E)$ and projecting with U_L gives, with $\mathsf{G} = U_\mathrm{L}(\nabla\mathsf{H})U$,

$$\mathsf{G} + \mathrm{diag}(E)\,X = X\,\mathrm{diag}(E) + \mathrm{diag}(\nabla E), \qquad X \equiv U_\mathrm{L}\nabla U. \tag{23}$$

Hence the energy gradients and the forces on each surface are

$$\nabla E_J = \left[U_{\mathrm{L}}(\nabla \mathsf{H})U\right]_{JJ}, \qquad \mathbf{F}_J = -\,\mathrm{Re}\;\nabla E_J. \tag{24}$$

Using $\nabla|a(\mathbf{R})\rangle = \sum_b |b\rangle D_{ba}$, the non-adiabatic coupling vectors (NACs) are

$$\boxed{\mathbf{d}_{JK} \equiv \langle\tilde{\Psi}_J|\nabla\Psi_K\rangle = \frac{\left[U_{\mathrm{L}}(\nabla \mathsf{H})U\right]_{JK}}{E_K - E_J} + \left[U_{\mathrm{L}}\,\mathsf{D}\,U\right]_{JK}, \qquad J \neq K,} \tag{25}$$

where $\langle\tilde{\Psi}_J|$ is the left (biorthogonal) state. The first term is the Hellmann–Feynman-like contribution from the $\mathbf{R}$-dependence of E_a, M_{ab} and Q_{ab}. The second term is the chain-rule contribution of the intrinsic electronic coupling $\langle g|\nabla e\rangle$, and it vanishes when $\mathbf{d} = 0$. Using Eq. (17), Eq. (25) is equivalent to $\mathbf{d}_{JK} = [U_{\mathrm{L}}\langle\cdot|\nabla\hat{H}|\cdot\rangle U]_{JK}/(E_K - E_J)$.

Properties:

- **Antisymmetry.** $U^T U = 1$ makes X antisymmetric, and D is real antisymmetric, so

$$\mathbf{d}_{JK} = -\mathbf{d}_{KJ}, \qquad \mathbf{d}_{JJ} = 0. \tag{26}$$

  For $\gamma > 0$ the NACs are complex and antisymmetric, but *not* anti-Hermitian.

- **Degeneracy guard.** If $|E_K - E_J| < 10^{-12}$, the first term of Eq. (25) is set to zero. Such pairs are exact crossings of uncoupled states, e.g. states of different JC manifolds.

- **Velocity projection.** The electronic equations need only the projection

$$(\mathbf{v}\cdot\mathbf{d})_{JK} = \frac{[U_{\mathrm{L}}(\mathbf{v}\cdot\nabla \mathsf{H})U]_{JK}}{E_K - E_J} + [U_{\mathrm{L}}(\mathbf{v}\cdot\mathsf{D})U]_{JK}, \tag{27}$$

  which is assembled directly from the contracted matrices $\mathbf{v}\cdot\nabla\mathsf{H}$ and $\mathbf{v}\cdot\mathsf{D}$.

6.2 Selection rule for the intrinsic term in the JC limit

D connects only $|g,n\rangle$ with $|e,n\rangle$, which have excitation numbers n and $n+1$. JC eigenstates have a definite N_{exc}, so $[U_{\mathrm{L}}\mathsf{D}U]_{JK} \neq 0$ only when $N_{\mathrm{exc}}(J) - N_{\mathrm{exc}}(K) = \pm 1$. In particular:

- the intrinsic coupling makes *no* contribution to $\mathbf{d}_{\mathrm{LP}_N,\mathrm{UP}_N}$ within one manifold;
- it couples $\mathrm{LP}_1/\mathrm{UP}_1$ to $|g,0\rangle$, and manifold N to manifold $N \pm 1$.

In the Rabi and PF models the counter-rotating and permanent-dipole terms mix the manifolds, so the intrinsic term does enter the LP–UP coupling. This is why the full NAC matrix is kept.

7 Analytical Jaynes–Cummings solution

$\hat{H}_{\mathrm{JC}}$ is block diagonal in $N = N_{\mathrm{exc}}$ (`AnalyticJCSolver`). The truncated basis splits into three groups:

- $N = 0$: the single state $|G_0\rangle = |g,0\rangle$, with $E = E_g$.
- $1 \le N \le n_{\mathrm{f}}$: the 2×2 blocks spanned by $\{|g,N\rangle, |e,N-1\rangle\}$,

$$\mathsf{H}_N = \begin{pmatrix} a_N & c_N \\ c_N & b_N \end{pmatrix}, \qquad a_N = E_g + N\tilde{\omega}, \quad b_N = E_e + (N-1)\tilde{\omega}, \quad c_N = \kappa M_{eg}\sqrt{N}. \tag{28}$$

- The uncoupled truncation state $|e,n_{\mathrm{f}}\rangle$, with $E = E_e + n_{\mathrm{f}}\tilde{\omega}$.

The states are ordered $[G_0, \mathrm{LP}_1, \mathrm{UP}_1, \ldots, \mathrm{LP}_{n_{\mathrm{f}}}, \mathrm{UP}_{n_{\mathrm{f}}}, |e,n_{\mathrm{f}}\rangle]$.

Energies. With the detuning $\delta = \omega - (E_e - E_g)$,

$$\Delta \equiv \frac{a_N - b_N}{2} = \frac{1}{2}\Big(\delta - \frac{i\gamma}{2}\Big), \qquad s_N = \sqrt{\Delta^2 + c_N^2}, \tag{29}$$

$$E_{\mathrm{UP}_N/\mathrm{LP}_N} = \frac{a_N + b_N}{2} \pm s_N = \frac{E_g + E_e + (2N-1)\tilde{\omega}}{2} \pm s_N. \tag{30}$$

Note that Δ does not depend on N. In the Hermitian limit $2s_N = \sqrt{\delta^2 + 4c_N^2}$, which gives the vacuum Rabi splitting $2\kappa|M_{eg}|\sqrt{N}$ at resonance. With loss at resonance, $2s_N = \sqrt{4c_N^2 - \gamma^2/4}$. This is real in the strong-coupling regime $c_N > \gamma/4$ and imaginary below it. The **exceptional point** is therefore

$$\delta = 0, \qquad \kappa|M_{eg}|\sqrt{N} = \gamma/4. \tag{31}$$

Eigenvectors. With the complex mixing angle θ_N defined by

$$\cos 2\theta_N = \frac{\Delta}{s_N}, \qquad \sin 2\theta_N = \frac{c_N}{s_N}, \tag{32}$$

the eigenvectors are

$$|\mathrm{UP}_N\rangle = \cos\theta_N |g, N\rangle + \sin\theta_N |e, N-1\rangle, \qquad |\mathrm{LP}_N\rangle = -\sin\theta_N |g, N\rangle + \cos\theta_N |e, N-1\rangle. \tag{33}$$

They are c-normalized automatically, because $\cos^2\theta + \sin^2\theta = 1$ also for complex θ. The half-angle functions are evaluated stably:

- if $|1 + \cos 2\theta| \geq |1 - \cos 2\theta|$, then $\cos\theta = \sqrt{(1 + \cos 2\theta)/2}$ and $\sin\theta = \sin 2\theta/(2\cos\theta)$;
- otherwise the roles of $\cos\theta$ and $\sin\theta$ are swapped.

The near-EP test uses $|\cos\theta|^2 + |\sin\theta|^2 > 10^3$.

Branch continuity. At $t = 0$ the principal branch of the square root is used ($\mathrm{Re}\, s_N \geq 0$). Afterwards the sign of s_N is chosen to minimize $|s_N - s_N^{\mathrm{prev}}|$, with $s_N^{\mathrm{prev}} = (E_{\mathrm{UP}_N} - E_{\mathrm{LP}_N})/2$ from the previous step. Without this, the principal branch would swap the labels when a trajectory crosses resonance below the EP. "UP" always denotes the $+s_N$ branch.

Gradients. With

$$\nabla\Delta = \tfrac{1}{2}(\nabla E_g - \nabla E_e), \qquad \nabla c_N = \kappa\sqrt{N}\,\nabla M_{eg}, \qquad \nabla s_N = \frac{\Delta\,\nabla\Delta + c_N \nabla c_N}{s_N}, \tag{34}$$

the energy gradients are

$$\nabla E_{\mathrm{UP}_N/\mathrm{LP}_N} = \tfrac{1}{2}(\nabla E_g + \nabla E_e) \pm \nabla s_N. \tag{35}$$

NACs. Within a block, the Hellmann–Feynman part is the gradient of the mixing angle,

$$\mathbf{d}^{\mathrm{HF}}_{\mathrm{LP}_N,\mathrm{UP}_N} = \nabla\theta_N = \frac{\Delta\,\nabla c_N - c_N\,\nabla\Delta}{2s_N^2} = -\mathbf{d}^{\mathrm{HF}}_{\mathrm{UP}_N,\mathrm{LP}_N}. \tag{36}$$

This is a Lorentzian in the detuning, of width set by c_N, and it diverges as s_N^{-2} at an EP. The HF part vanishes between different blocks.

The intrinsic part $[U^T \mathsf{D} U]$ couples only neighbouring manifolds. For X_N in block N and Y_{N+1} in block $N+1$,

$$[U^T \mathsf{D} U]_{X_N, Y_{N+1}} = \mathbf{d}\, \langle g, N | X_N \rangle \, \langle e, N | Y_{N+1} \rangle, \tag{37}$$

where $\langle \cdot | \cdot \rangle$ here denotes the component of the c-normalized column. Explicitly:

$$[\,\cdot\,]_{\mathrm{UP}_N, \mathrm{UP}_{N+1}} = \mathbf{d} \cos\theta_N \sin\theta_{N+1}, \qquad [\,\cdot\,]_{\mathrm{UP}_N, \mathrm{LP}_{N+1}} = \mathbf{d} \cos\theta_N \cos\theta_{N+1}, \tag{38}$$

$$[\,\cdot\,]_{\mathrm{LP}_N, \mathrm{UP}_{N+1}} = -\mathbf{d} \sin\theta_N \sin\theta_{N+1}, \qquad [\,\cdot\,]_{\mathrm{LP}_N, \mathrm{LP}_{N+1}} = -\mathbf{d} \sin\theta_N \cos\theta_{N+1}, \tag{39}$$

with G_0 treated as block 0 with $\cos\theta_0 = 1$, so that, e.g., $[\,\cdot\,]_{G_0, \mathrm{UP}_1} = \mathbf{d} \sin\theta_1$ and $[\,\cdot\,]_{G_0, \mathrm{LP}_1} = \mathbf{d} \cos\theta_1$. The truncation state $|e, n_\mathrm{f}\rangle$ acts as block $n_\mathrm{f} + 1$ with unit component on $|e, n_\mathrm{f}\rangle$. These expressions hold in the analytic gauge. The code afterwards applies the sign convention of Sec. 5.2, which multiplies $\mathbf{d}_{JK}$ by $s_J s_K$ with $s_J = \pm 1$.

The analytic and numerical solvers agree to 10^{-10} in energies, gradients and NACs (test suite).

8 Nuclear dynamics: velocity Verlet and Langevin (BBK)

The nuclei move classically on the active surface a, with the force of Eq. (24). An optional Langevin bath with friction ζ (in inverse time) and temperature $k_B T$ models vibrational relaxation and thermalization by the environment. It acts uniformly on all degrees of freedom and is integrated with the Brünger–Brooks–Karplus scheme:

$$\mathbf{P}_{1/2} = \mathbf{P}(t) + \frac{\Delta t}{2}\Big[\mathbf{F}_a(\mathbf{R}(t)) - \zeta \mathbf{P}(t) + \boldsymbol{\eta}\Big], \tag{40}$$

$$\mathbf{R}(t + \Delta t) = \mathbf{R}(t) + \Delta t\, \mathsf{M}^{-1} \mathbf{P}_{1/2}, \tag{41}$$

$$\mathbf{P}(t + \Delta t) = \frac{\mathbf{P}_{1/2} + \frac{\Delta t}{2}\big[\mathbf{F}_a(\mathbf{R}(t + \Delta t)) + \boldsymbol{\eta}'\big]}{1 + \zeta \Delta t / 2}, \tag{42}$$

with independent Gaussian random forces satisfying the fluctuation–dissipation relation

$$\langle \eta_i \eta_j \rangle = \frac{2 \zeta M_i k_B T}{\Delta t}\, \delta_{ij}. \tag{43}$$

Three limits follow:

- $\zeta = 0$ gives velocity Verlet.
- $T = 0$ with $\zeta > 0$ gives deterministic damping (used in the legacy example).
- $T > 0$ gives canonical sampling of the nuclei on the active surface.

The legacy code applied the friction twice in the second half-step (an effective 1.5ζ); that error is not present here.

9 Electronic propagation

The electronic–photonic wavefunction is

$$|\Psi(t)\rangle = \sum_{a,n} \psi_{an}(t) |a(\mathbf{R}(t))\rangle \otimes |n\rangle. \tag{44}$$

It is propagated in the product basis rather than the polaritonic basis. This avoids gauge and trivial-crossing problems of the polaritonic basis at narrow LP/UP avoided crossings. Inserting it into $i\partial_t\Psi = \hat{H}\Psi$ along the classical path gives

$$\dot{\psi} = \mathsf{K}\,\psi, \qquad \mathsf{K}(t) = -i\,\mathsf{H}(\mathbf{R}(t)) - \mathbf{v}(t)\cdot\mathsf{D}(\mathbf{R}(t)). \tag{45}$$

- $\mathbf{v}\cdot\mathsf{D}$ is real antisymmetric and generates a rotation.
- The only norm-changing term is the anti-Hermitian photon part:

$$\frac{d}{dt}\|\psi\|^2 = -\gamma\,\langle\psi|\hat{1}\otimes\hat{b}^\dagger\hat{b}|\psi\rangle. \tag{46}$$

- In the polaritonic representation, Eq. (45) is the familiar $\dot{c}_J = -iE_Jc_J - \sum_K(\mathbf{v}\cdot\mathbf{d})_{JK}c_K$.

Each nuclear step $[t, t+\Delta t]$ is divided into n_s substeps of length $h = \Delta t/n_s$, evaluated at the midpoints $x_j = (j+\frac{1}{2})/n_s$. Within the step, H, $\nabla\mathsf{H}$, D and $\mathbf{v}$ are interpolated linearly between the two endpoint geometries:

$$\mathsf{H}(x) = (1-x)\mathsf{H}(t) + x\mathsf{H}(t+\Delta t),\ \text{etc.}, \qquad \psi \leftarrow \exp\big[h\,\mathsf{K}(x_j)\big]\,\psi. \tag{47}$$

The matrix exponential is evaluated exactly with `scipy.linalg.expm`. The endpoint velocities are taken before any hop.

10 Fewest-switches hopping

10.1 Hopping probabilities

At every substep the interpolated $\mathsf{H}(x_j)$ is diagonalized with the tracking of Sec. 5.2 (no extra model calls). The amplitudes $c = U_{\mathrm{L}}\psi$ and the matrix $\mathbf{v}\cdot\mathbf{d}$ of Eq. (27) are formed, and Tully's flux out of the active state a into each other state k is accumulated:

$$g_{a\to k} \mathrel{+}= h\,\frac{\max\big(0,\ -2\,\mathrm{Re}\,\big[c_k^*\,(\mathbf{v}\cdot\mathbf{d})_{ka}\,c_a\big]\big)}{|c_a|^2}, \qquad k\neq a, \tag{48}$$

skipped when $|c_a|^2 < 10^{-14}$. Both upward and downward hops are allowed.

10.2 Hop selection and momentum rescaling

If no photon jump occurred in the step (Sec. 11), a uniform random number $\xi\in[0,1)$ is compared with the cumulative sums of $g_{a\to k}$, and a hop $a\to k$ is attempted at the end-of-step geometry. Energy is conserved with respect to the real parts of the surfaces, $\Delta E = \mathrm{Re}(E_k - E_a)$. The rescaling direction $\mathbf{u}$ is one of the following:

- **NAC** (default): the complex vector $\mathbf{d}_{ak}$ rotated by a global phase to make it as real as possible,

$$\eta = \tfrac{1}{2}\arg\sum_i d_{ak,i}^2, \qquad \mathbf{u} = \mathrm{Re}\left(\mathbf{d}_{ak}e^{-i\eta}\right), \tag{49}$$

  which maximizes $\|\,\mathrm{Re}(\mathbf{d}_{ak}e^{-i\eta})\|$. In 1-D this reduces to rescaling along $\pm\hat{R}$.

- **velocity**: $\mathbf{u} = \mathbf{v}$;

- **none**: no rescaling (a test option; it does not conserve energy).

With $A = \sum_i u_i^2/2M_i$ and $B = \sum_i P_i u_i/M_i$, energy conservation requires $A\alpha^2 - B\alpha + \Delta E = 0$:

$$\mathbf{P} \leftarrow \mathbf{P} - \alpha\,\mathbf{u}, \qquad \alpha = \frac{B - \mathrm{sgn}(B)\sqrt{B^2 - 4A\Delta E}}{2A}, \tag{50}$$

which is the smaller-magnitude root. If $B^2 - 4A\Delta E < 0$ the hop is **frustrated**. The active state is kept, and either $\mathbf{P}$ is left unchanged (`"keep"`) or its component along $\mathbf{u}$ is reversed, $\mathbf{P} \leftarrow \mathbf{P} - (B/A)\mathbf{u}$ (`"reverse"`).

10.3 Decoherence correction (optional)

With constant C (typically $0.1\,E_h$), the energy-based decoherence correction of Granucci and Persico damps the inactive amplitudes after each step:

$$\tau_k = \frac{1 + C/T_{\mathrm{kin}}}{|\,\mathrm{Re}(E_k - E_a)|}, \qquad c_k \leftarrow c_k\, e^{-\Delta t/\tau_k}\ (k \neq a), \qquad |c_a| \leftarrow \Big(\textstyle\sum_K |c_K|^2_{\mathrm{before}} - \sum_{k\neq a} |c_k|^2\Big)^{1/2}. \tag{51}$$

Then $\psi \leftarrow Uc$, rescaled to its previous norm. The correction is off by default.

11 Photon loss as quantum jumps

Cavity loss is described by the Lindblad equation

$$\dot\rho = -i[\hat H_{\mathrm{r}}, \rho] + \gamma\Big(\hat b\rho\hat b^\dagger - \tfrac{1}{2}\{\hat b^\dagger\hat b, \rho\}\Big), \tag{52}$$

where $\hat H_{\mathrm{r}}$ is $\hat H$ with $\tilde\omega \to \omega$. It is unravelled with the Monte Carlo wavefunction method. The no-jump evolution is generated by the effective Hamiltonian $\hat H_{\mathrm{r}} - \frac{i}{2}\gamma\hat b^\dagger\hat b$, which is exactly the complex-frequency Hamiltonian of Sec. 4. After the electronic propagation of each nuclear step:

1. Compute the jump probability

$$p_{\mathrm{jump}} = 1 - \|\psi\|^2 \approx \gamma\,\Delta t\,\langle\hat b^\dagger\hat b\rangle, \tag{53}$$

   where ψ was normalized at the start of the step.

2. With probability p_{jump} a photon is emitted:

$$\psi \leftarrow \frac{(\hat 1 \otimes \hat b)\psi}{\|(\hat 1 \otimes \hat b)\psi\|}, \tag{54}$$

   and the new active surface is drawn from ρ_J of the post-jump state (with $c = U_{\mathrm{L}}\psi$ at the current geometry). The momentum is *not* rescaled: the emitted photon carries away the energy $\approx \mathrm{Re}(E_{\mathrm{old}} - E_{\mathrm{new}}) \approx \omega$. No hop is attempted in a step with a jump.

3. Otherwise, renormalize: $\psi \leftarrow \psi/\|\psi\|$.

Because $\hat b$ lowers the photon number by one, a jump from the N-excitation manifold lands in the $(N-1)$-manifold. For example, a jump from UP_2 lands in $\mathrm{LP}_1/\mathrm{UP}_1$, not in G_0. This is the generalization to arbitrary n_{f} of the legacy "ground-state sink", which is correct only for $n_{\mathrm{f}} = 1$. With `photon_loss=False`, ψ is never renormalized and $\|\psi\|^2$ is the survival probability; this is a diagnostic mode.

The total energy $T_{\mathrm{kin}} + \mathrm{Re}\,E_a$ is conserved between jumps (to $\lesssim 10^{-6}E_h$ in the tests). It drops by the photon energy at each jump.

12 Initial conditions

Nuclei. The nuclear positions and momenta are sampled from the ground-state Wigner distribution of a harmonic reference. For a 1-D well with force constant k and $a = \sqrt{kM} = M\omega_0$,

$$R \sim \mathcal{N}\Big(R_0, \frac{1}{2a}\Big), \qquad P \sim \mathcal{N}\Big(0, \frac{a}{2}\Big) \qquad \text{(variances)}. \tag{55}$$

The legacy code used the momentum width $\sqrt{a - \sqrt{\pi a}/2}$ in place of $\sqrt{a/2}$, about 1.39 times too broad for its parameters.

For many atoms, the Cartesian Hessian $\mathcal{H}$ at the minimum $\mathbf{R}_0$ is mass-weighted, $\mathsf{M}^{-1/2}\mathcal{H}\mathsf{M}^{-1/2} = L\,\mathrm{diag}(\omega_i^2)L^T$, and each normal mode is sampled independently:

$$q_i \sim \mathcal{N}\Big(0, \frac{1}{2\omega_i}\Big), \quad p_i \sim \mathcal{N}\Big(0, \frac{\omega_i}{2}\Big), \qquad \mathbf{R} = \mathbf{R}_0 + \mathsf{M}^{-1/2}Lq, \quad \mathbf{P} = \mathsf{M}^{1/2}Lp. \tag{56}$$

Modes with ω_i below a cutoff (translations and rotations) are left at zero.

Electrons and photons. The trajectory starts on a chosen polaritonic state a. The initial wavefunction is that state's right eigenvector, normalized in the ordinary sense: $\psi(0) = u_a/\|u_a\|$.

Random numbers. Each trajectory gets an independent `numpy` generator spawned from a single `SeedSequence`, so ensembles are reproducible and the streams do not overlap.

13 Algorithm summary

Algorithm 1 One nuclear step of `FSSH.step`

$\mathbf{P}_{1/2} \leftarrow \mathbf{P} + \frac{\Delta t}{2}(\mathbf{F}_a - \zeta\mathbf{P} + \boldsymbol{\eta})$; $\quad \mathbf{R}' \leftarrow \mathbf{R} + \Delta t\,\mathsf{M}^{-1}\mathbf{P}_{1/2}$
Evaluate the electronic model at $\mathbf{R}'$; align the phase of $|e\rangle$ (Eq. 6); project on $\hat{\boldsymbol{\epsilon}}$
Build $\mathsf{H}, \nabla\mathsf{H}, \mathsf{D}$ at $\mathbf{R}'$; diagonalize with tracking; compute ∇E_J and $\mathbf{d}_{JK}$ (Eqs. 24, 25)
$\mathbf{P}' \leftarrow [\mathbf{P}_{1/2} + \frac{\Delta t}{2}(\mathbf{F}_a(\mathbf{R}') + \boldsymbol{\eta}')]/(1 + \zeta\Delta t/2)$
for $j = 0, \ldots, n_s - 1$ **do** ▷ electronic substeps
    Interpolate $\mathsf{H}, \nabla\mathsf{H}, \mathsf{D}, \mathbf{v}$ at x_j; diagonalize; $c \leftarrow U_{\mathrm{L}}\psi$; accumulate $g_{a\to k}$ (Eq. 48)
    $\psi \leftarrow \exp[h(-i\mathsf{H} - \mathbf{v}\cdot\mathsf{D})]\,\psi$
end for
if photon loss **and** $\xi_1 < 1 - \|\psi\|^2$ **then**
    $\psi \leftarrow \hat{b}\psi/\|\hat{b}\psi\|$; draw new a from $|c_J|^2$ ▷ quantum jump, no rescaling
else
    $\psi \leftarrow \psi/\|\psi\|$; select a hop from $g_{a\to k}$ with ξ_2; rescale $\mathbf{P}'$ or treat it as frustrated
end if
Optional decoherence correction

Recorded observables. Every `stride` steps the code records:

- t, $\mathbf{R}$, $\mathbf{P}$ and the active state;
- $\mathrm{Re}\,E_J$ for all states, and the populations ρ_J;

- $\|\psi\|^2$ and the photon number $\langle\psi|\hat{b}^\dagger\hat{b}|\psi\rangle/\langle\psi|\psi\rangle$;
- T_{kin} and $T_{\mathrm{kin}} + \mathrm{Re}\, E_a$;
- the list of hop, frustrated-hop and jump events.

`scan_pes` returns, along a path of geometries, the surfaces, their slopes and the NACs projected on the path tangent, and the photon content of each state.

14 Approximations, open choices, and physics not yet included

14.1 Approximations built into the current formalism

1. **Two-level truncation of Pauli–Fierz.** The truncated length-gauge PF Hamiltonian is not gauge invariant. The projected DSE $(M^2)_{ab}$ also differs from $\langle a|(\hat{\boldsymbol{\epsilon}}\cdot\hat{\boldsymbol{\mu}})^2|b\rangle$, which includes the virtual excited states. The `dipole_squared` hook allows the exact projected second moment to be used if the ML model can provide it.
2. **Dipole approximation and a single mode.** λ is uniform over the molecule, and only one mode and one polarization are treated.
3. **Single molecule.** No collective (Tavis–Cummings) effects, no dark states, no disorder.
4. **Non-Hermitian surfaces.** The forces use $\mathrm{Re}\,\nabla E_J$. The populations rely on the biorthogonal amplitudes, which are non-orthogonal when $\gamma > 0$, and the formalism breaks down near exceptional points (only detected, not treated).
5. **FSSH.** Classical nuclei, so there is no tunnelling, and zero-point energy may leak. The decoherence correction is optional and phenomenological. The rescaling direction for complex NACs is a choice.
6. **Langevin bath.** The friction is uniform, isotropic and Markovian, and acts on all nuclear degrees of freedom. It is a phenomenological stand-in for the solvent or the rest of the molecule.
7. **Zero-temperature photon bath.** Only emission ($\hat{b}$ jumps) is included. There is no thermal photon absorption ($\hat{b}^\dagger$ jumps) and no pumping.

14.2 Physics that may need to be added for the ML application

1. **Intrinsic NAC from the ML model.** The formalism uses $\langle g|\nabla e\rangle$ fully (Eq. 25), but most ML models do not predict it, and it defaults to zero. The model would need to learn it with a phase consistent with $\boldsymbol{\mu}_{eg}$, or it could be estimated from $\nabla\mathsf{H}$ of an underlying diabatic model.
2. **Molecular radiative and non-radiative decay, and pure dephasing.** Only cavity loss is included. Additional Lindblad channels ($\hat{\sigma}_-$ emission into free space, $\hat{\sigma}_z$ dephasing) can be added as further jump operators.
3. **Thermal photon occupation** at finite cavity temperature ($\hat{b}^\dagger$ jumps with rate $\gamma\bar{n}$).
4. **Several modes or polarizations,** and orientational averaging over $\hat{\boldsymbol{\epsilon}}$ for isotropic samples.

5. **More than two electronic states** (e.g. S_2, triplets) when the ML model provides them. The product-basis construction generalizes directly.

6. **Geometry dependence of λ,** or non-dipole effects, for plasmonic nanocavities.

7. **Initial excitation protocol.** Starting on UP/LP assumes a prepared polariton. A pulse-driven preparation, or initialization from a transition-dipole-weighted (Franck–Condon) excitation, may be more realistic.

A Map from equations to code

Content	Module / function
Electronic data, finite differences, phase alignment (Sec. 2)	`electronic.py`: `CallableModel`, `align_phase`
Projection on $\hat{\boldsymbol{\epsilon}}$, cavity parameters (Sec. 3)	`cavity.py`: `CavityMode`, `project`
H, $\nabla\mathsf{H}$, D (Sec. 4)	`hamiltonian.py`: `CavityHamiltonian`
c-normalization, tracking, NACs (Secs. 5–6)	`polariton.py`: `diagonalize`, `NumericalSolver`
Analytical JC (Sec. 7)	`polariton.py`: `AnalyticJCSolver`
Nuclear and electronic propagation, hops, jumps (Secs. 8–11)	`fssh.py`: `FSSH.step`
Initial conditions (Sec. 12)	`sampling.py`
Surface scans, trajectory output	`io.py`: `scan_pes`, `TrajectoryRecord`
Reference models	`models.py`: `DoubleWellModel`, `DiabaticTwoSta`